Investigating the Early Signatures of Feature Learning in Deep Linear Networks

Shaunak Bhandarkar[1]*, Boris Hanin[1] *Correspondence: shaunak@princeton.edu

[1]Princeton University

Motivation

Deep neural networks, unlike **kernel models**, learn data-dependent features in their internal layers [1], a capacity widely credited for their empirical success.

$$z(x;\theta) = \sum_{j=1}^{N} W_j \phi_j(x)$$

Kernel model: **fix** the ϕ_j's

Deep neural network: **learn** the ϕ_j's

Question: what data-dependent features does a deep network learn, and when do they generalize better than fixed ones? Two issues stand in the way:

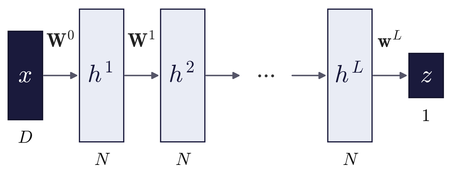
(i) Learning dynamics are nonlinear and non-convex, making it hard to pin down learned features in closed-form.

(ii) Whether learned features are useful depends on the *data distribution*, which need not be isotropic and often possesses low-dimensional structure [2].

Here, we address these issues with three modeling choices: studying deep *linear* networks, a *single* gradient (GD) step, and a *spiked* data distribution.

Setup

Model. An L-layer *linear* network with width N [3]. Dataset: $X \in \mathbb{R}^{D\times n}$, $y \in \mathbb{R}^n$.



$$z(x;\theta) = \frac{1}{N}\mathbf{w}^L \cdot h^L(x;\theta)$$

$$h^{\ell+1}(x;\theta) = \frac{1}{\sqrt{N}}\mathbf{W}^\ell h^\ell(x;\theta)$$

$$h^1(x;\theta) = \frac{1}{\sqrt{D}}\mathbf{W}^0 x$$

$$\Phi^\ell_{t;ij} := \frac{h^\ell(x_i;\theta_t)\cdot h^\ell(x_j;\theta_t)}{N}$$

Layer-ℓ feature kernel

$$\Phi^0 := K_x = \frac{1}{D}X^T X \in \mathbb{R}^{n\times n}$$

Baseline input kernel

Training. *One* GD step [4] on the MSE loss $\mathscr{L} = \frac{1}{2n}\sum_{i=1}^{n}\left(z(x_i;\theta) - y_i\right)^2$.

$$\mathbf{W}^\ell_{t=1} = \mathbf{W}^\ell_{t=0} + \frac{\eta}{n\sqrt{N_\ell}}\sum_{i=1}^{n}\left(y_i - z(x_i;\theta_0)\right) g^{\ell+1} h(x_i;\theta_0)^{\ell T}$$

Here, η is the learning rate and $g^\ell = N\frac{\partial z}{\partial h^\ell}$ is a backpropagated gradient.

Data. Linear regression on a *spiked matrix model* [5], with **spike strength** $\gamma \geq 0$ and **spike-target alignment** $\rho \in [-1,1]$. Here, $\gamma = 0$ means isotropic inputs and $\rho = 0$ means the spike is orthogonal to the target.

$$x_i \overset{\text{iid}}{\sim} \mathcal{N}(0, I_D + \gamma v v^T) \qquad y_i = \beta_\star^T x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,\sigma^2)$$

Here, $\beta_\star, v \in \mathbb{S}^{D-1}$ are fixed so that $\beta_\star^T v = \rho$. We study the overparameterized proportional limit $n, D \to \infty$, $n/D \to \psi \in (0,1)$ [6], with $D > n+1$.

Central Questions

(**Q1**) For arbitrary data $\{(x_i, y_i)\}_{i=1}^n$ how do $\Phi^\ell_{t=0}$ and $\Phi^\ell_{t=1}$ differ as $N \to \infty$?

(**Q2**) For spiked data, when does *kernel ridge regression* (KRR) on the one-step kernel lower generalization error in the proportional limit as $n/D \to \psi \in (0,1)$?

Punchline: one step gives each layer's feature kernel a target-aligned low-rank spike, setting it up to improve generalization for data distributions with task-aligned low-rank structure, and to fail to do so if this structure is missing.

Result 1: The feature kernel update

One step imprints a target-aligned rank-one spike on every feature kernel, amplified by learning rate and depth.

$$\Phi^\ell_{t=0} \xrightarrow{L^2} K_x \text{ and } \Phi^\ell_{t=1} \xrightarrow{L^2} K^\ell_{feat} := K_x + \frac{\eta^2 d_\ell}{n^2} K_x y y^T K_x$$

where $d_\ell := \frac{\ell(\ell+1)(2\ell+1)}{6}$. This update grows with the learning rate η **and** the layer ℓ. Feature evolution after one step is more dramatic in deeper layers.

Result 2: The effect on kernel estimation

One step of feature learning uniformly inflates estimation.

For a kernel K and ridge $\lambda \geq 0$, $\hat{f}_\lambda(x) = K(x,X)\big(K(X,X)+\lambda I_n\big)^{-1} y$ is the estimator obtained by KRR. Using KRR on K^ℓ_{feat} and K_x, we find:

$$\hat{f}_{\lambda;\text{feat}}(x) = c_\lambda \hat{f}_{\lambda;\text{init}}(x)$$

$$c_\lambda := 1 + \frac{\lambda k_\ell \frac{D}{n^2} y^T K_x (K_x+\lambda I_n)^{-1} y}{1 + k_\ell \frac{D}{n^2} y^T K_x (K_x+\lambda I_n)^{-1} K_x y} \geq 1 \qquad k_\ell := \frac{(\eta\ell)^2}{D}$$

1. $c_\lambda = 1$ when $\lambda = 0$: **one step of feature learning is invisible without ridge.**
2. Asymptotically, $c_\lambda > 1$ only once $k_\ell \asymp 1$. **Thus, k_ℓ sets the scale of feature learning and produces nontrivial inflation if $\eta\ell \asymp \sqrt{D}$.**

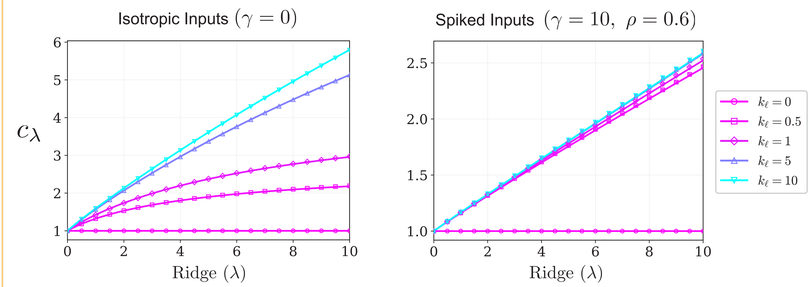


Fig. 1. Asymptotic values of c_λ across ridge strengths and for different k_ℓ. Here, $\psi = \sigma = 0.5$. Lines show theory; markers show simulations ($D = 1000$, $n = 500$).

Result 3: The tradeoff between ridge and inflation

Ridge and inflation oppose each other on the bias-variance axis.

We derive asymptotic formulas for the bias $\tilde{B}$, variance $\tilde{V}$, and generalization error $\tilde{G}$ of any KRR estimator $\hat{f}_{\lambda;k_\ell}$. $k_\ell = 0$ recovers $\hat{f}_{\lambda;\text{init}}$. Around $\lambda = 0$ we find:

$$\partial_\lambda \tilde{B}(\hat{f}_{\lambda;k_\ell})\Big|_{\lambda=0} < \partial_\lambda \tilde{B}(\hat{f}_{\lambda;0})\Big|_{\lambda=0}, \quad \partial_\lambda \tilde{V}(\hat{f}_{\lambda;k_\ell})\Big|_{\lambda=0} > \partial_\lambda \tilde{V}(\hat{f}_{\lambda;0})\Big|_{\lambda=0} \quad \forall k_\ell > 0$$

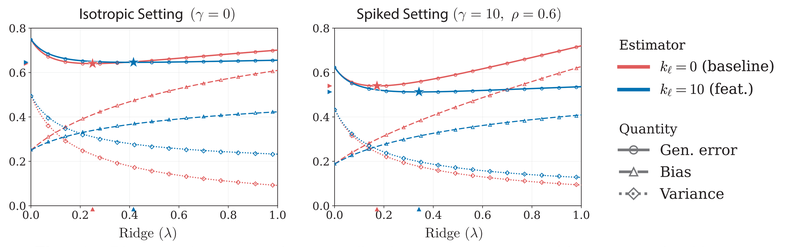


Fig. 2. Bias, variance, and generalization error across ridge strength. Here, $\psi = \sigma = 0.5$. Lines show theory; markers show simulations ($D = 1000$, $n = 500$).

Ridge **shrinks** the ridgeless interpolator $\hat{f}_{0;0}$, lowering variance but raising bias. **Inflation** acts as a corrective measure for bias but also amplifies noise (variance).

Result 4: Identifying when features are useful

A feature learning advantage (FLA) exists when the data possesses a sufficiently strong, task-aligned spike, and vanishes in the absence of a spike or task-alignment.

FLA: when some one-step estimators can outperform *any* baseline estimator, i.e.

$$\inf_{\lambda\geq 0, k_\ell > 0} \tilde{G}\left(\hat{f}_{\lambda;k_\ell}\right) < \inf_{\lambda \geq 0} \tilde{G}\left(\hat{f}_{\lambda;0}\right).$$

Sufficient condition. Suppose $\gamma, \sigma^2 > 0$, and

$$\frac{(1+\gamma\psi)^2}{\gamma(1+\gamma)(1-\psi)^2} < \frac{\rho^2}{\sigma^2} < \frac{(1+\gamma\psi)^3}{\gamma(1+\gamma)(1-\psi)^3}.$$

Then, a feature learning advantage holds. **Here, ρ^2/σ^2 is an effective signal-to-noise ratio that produces an FLA when it is sufficiently large.**

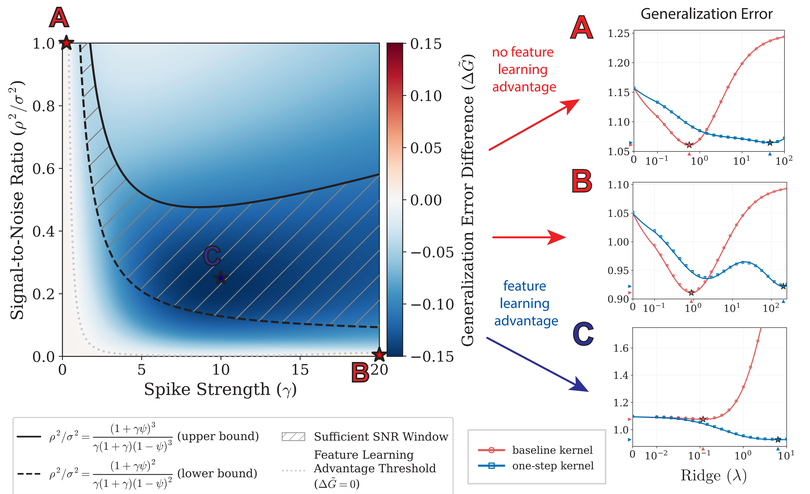


Fig. 3. Comparison of minimal generalization error over λ between one-step ($k_\ell = 10$) and baseline estimators. Blue denotes an FLA. Here, $\psi = 0.2$ and $\sigma^2 = 1$. To the right, lines show theory and markers show simulations ($D = 5000$, $n = 1000$).

A key non-result. Suppose $\gamma = 0$ or $\rho = 0$. Then, for any $\lambda, k_\ell \geq 0$

$$\tilde{G}(\hat{f}_{\lambda;k_\ell}) \geq \tilde{G}(\hat{f}_{\sigma^2;0}).$$

In particular, no feature learning estimator can beat the optimally tuned ridge baseline ($\lambda^\star = \sigma^2$).

Conclusions

Our work suggests that GD is well-suited to learning structured low-dimensional information embedded in high-dimensional data, even early in training. In future work, we seek to extend our analyses to the following settings:

- Deep *nonlinear* neural networks
- Training across *multiple* gradient steps
- Data distributions with *multiple* "spikes"
- Training with other learning rules, or with more complex optimizers

References

1. Damian, A., Lee, J., & Soltanolkotabi, M. (2022). Neural networks can learn representations with gradient descent. COLT.
2. Ba, J., Erdogdu, M., Suzuki, T., Wang, Z., & Wu, D. (2023). Learning in the presence of low-dimensional structure: A spiked random matrix perspective. NeurIPS.
3. Bordelon, B. & Pehlevan, C. (2022). Self-consistent dynamical field theory of kernel evolution in wide neural networks. NeurIPS.
4. Ba, J., Erdogdu, M., Suzuki, T., Wang, Z., Wu, D., & Yang, G. (2022). High-dimensional asymptotics of feature learning: How one gradient step improves the representation. NeurIPS.
5. Johnstone, I. M. (2001). On the distribution of the largest eigenvalue in principal components analysis. The Annals of Statistics.
6. Hastie, T., Montanari, A., Rosset, S., & Tibshirani, R. J. (2022). *Surprises in high-dimensional ridgeless least squares interpolation*. The Annals of Statistics.